FIBRE PRODUCTS OF LIMIT PRESERVING TRANSFORMATIONS ARE LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 478-503, following the statement at 469-476.

Editorial note. The trailing 'Some details omitted.' covers only the mechanical verification of the displayed set-theoretic identity (three-line element chase); flagged in the notes per the b68 standard-substitution precedent.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 502 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Characterization of limit preservation. lemma-characterize-relative-limit-preserving (same chapter, card L26322, cited at ~478-480). Characterization (2) reduces the check to T = lim T_i.

Source reference. Original lemma 06BC: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 351–351. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $E, F, G, H : (Sch/S)^{opp}_{fppf} \to Sets$. Let $a : F \to G$, $b : H \to G$, and $c : G \to E$ be transformations of functors. If c, $c \circ a$, and $c \circ b$ are limit preserving, then $F \times_G H \to E$ is too.*

Proof. Let $T = \lim_{i \in I} T_i$ as in characterization (2) of Lemma 06BC. Then we have

$$\operatorname{colim}(F \times_G H)(T_i) = \operatorname{colim} F(T_i) \times_{\operatorname{colim} G(T_i)} \operatorname{colim} H(T_i)$$

as filtered colimits commute with finite products. Our goal is thus to show that

$$\begin{array}{ccc} \operatorname{colim} F(T_i) \times_{\operatorname{colim} G(T_i)} \operatorname{colim} H(T_i) & \longrightarrow & F(T) \times_{G(T)} H(T) \\ \downarrow & & \downarrow \\ \operatorname{colim}_i E(T_i) & \longrightarrow & E(T) \end{array}$$

is a fibre product diagram. This follows from the observation that given maps of sets $E' \to E$, $F \to G$, $H \to G$, and $G \to E$ we have

$$E' \times_E (F \times_G H) = (E' \times_E F) \times_{(E' \times_E G)} (E' \times_E H)$$

Some details omitted. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.